

Fractional Integral Terminal Sliding Mode Control for Servo Motor Systems in Phenotyping Robot in Precision Agriculture

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Abstract: This paper presents a fractional integral terminal sliding mode control (FITSMC) for speed tracking of a DC motor employed in phenotyping robots for green environment applications. Phenotyping robots require high-precision motion control to ensure reliable and repeatable measurements of plants under field conditions where mechanical and load disturbances exist. To address these challenges, the proposed FITSMC ensured finite-time convergence of speed-tracking errors, even in the presence of unknown disturbances and modeling uncertainties. Unlike conventional sliding mode control, terminal sliding mode control (TSMC) enables faster and more accurate convergence without requiring infinite control effort near the origin. The stability is analytically proven using the Lyapunov theory. Simulation results confirm the effectiveness and robustness of the proposed method, highlighting its potential for deployment in green robotic systems for precision agricultural applications.

Keywords: DC Motor, Finite-Time Speed Tracking, Fractional Integral Terminal Sliding Mode Control (FITSMC), Phenotyping Robots.

1. INTRODUCTION

Plant phenotyping is a critical component of modern precision agriculture, where the accurate measurement of plant traits such as height, leaf area, and canopy structure plays a pivotal role in crop breeding and stress analysis [1]. Phenotyping robots—platforms equipped with sensors and actuators—have emerged as a powerful solution to automate and accelerate the phenotyping process in both greenhouse and field environments. A key requirement for such robots is the ability to execute precise and stable motion trajectories despite the presence of disturbances, nonlinear dynamics, and uncertain operating conditions.

DC motors are widely used as actuators in phenotyping robots due to their simplicity, cost-effectiveness, and ease of control. However, the dynamic behavior of DC motors is often affected by modeling inaccuracies, load variations, frictional forces, and unmeasured disturbances from the operating environment. These uncertainties can degrade control performance and compromise the reliability of phenotyping data.

To address these challenges, sliding mode control (SMC) has been extensively explored due to its robustness and insensitivity to uncertainties [2], [3]. Nevertheless, conventional SMC may exhibit slow convergence. To overcome this drawback, fractional integral terminal sliding mode control (FITSMC) has been proposed as an advanced alternative. FITSMC [4] combines integral sliding mode control (ISMC) and terminal sliding mode control (TSMC), with a fractional exponential order introduced in the error term of the integral sliding surface. Since the DC motor model for speed control has a relative degree of one, ISMC is chosen [4]. Moreover, TSMC introduces a nonlinear sliding surface that ensures



Fig. 1 Phenotyping robot designed for precision agriculture, developed by our research group.

finite-time convergence, providing faster response and enhanced precision compared to conventional SMC while maintaining robustness against system uncertainties.

In this study, we investigate the application of FITSMC for DC speed motor control in a phenotyping robot. A system identification process was conducted on a real DC motor, and the identified model was used to simulate the proposed control strategy. The robotic platform architecture, including the mechanical design, was developed by our research team, with a focus on field deployment for plant phenotyping robots. Although real-time implementation is left for future work, simulation results demonstrate the effectiveness of the FITSMC in achieving accurate tracking and robustness to disturbances, confirming its potential for deployment in agricultural robotics.

This article is organized as follows: Section 2 introduces preliminaries on the DC motor model and finite-time control. The FITSMC Design is proposed in Section 3. In Section 4, the DC motor, its hardware, and system identification are presented. The simulation and results are discussed in Section 5. Section 6 concludes the paper.

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2. PRELIMINARIES

2.1. DC Motor Model

2.1.1. Transfer Function Model

The behavior of a PMDC motor is governed by four equations: electrical, back electromotive force (EMF) voltage, torque generation, and mechanical equations [5]. These equations are given by:

$$v_a = R_a i_a + L_a \frac{di_a}{dt} + e_a \quad (1)$$

$$e_a = K_b \omega_m \quad (2)$$

$$T_m = J_m \frac{d\omega_m}{dt} + D_m \omega_m + T_L \quad (3)$$

$$T_m = K_t i_a \quad (4)$$

where v_a denotes the input armature voltage (V), e_a denotes the back EMF voltage (V), i_a is the armature current (A), ω_m represents the angular velocity (rad/s), T_m is the electromagnetic torque (N·m), R_a is the armature resistance (Ω), L_a is the inductance of the armature (H), J_m is the moment of inertia ($\text{kg}\cdot\text{m}^2$), D_m is the viscous damping coefficient ($\text{N}\cdot\text{m}\cdot\text{s}/\text{rad}$), K_t is the torque constant ($\text{N}\cdot\text{m}/\text{A}$), and K_b is the back EMF constant ($\text{V}\cdot\text{s}/\text{rad}$),

The transfer function from the input voltage to the angular velocity is expressed by:

$$\frac{\omega_m(s)}{V_a(s)} = \frac{K_t}{(L_a s + R_a)(J_m s + D_m) + K_t K_b} \quad (5)$$

Since the electrical time constant L_a/R_a is much smaller than the mechanical time constant J_m/D_m , the armature inductance L_a can be neglected. The transfer function in Eq. (5) is simplified to:

$$\frac{\omega_m(s)}{V_a(s)} = \frac{K_t}{(R_a J_m) s + (R_a D_m + K_t K_b)} \quad (6)$$

2.1.2. State-Space Model

The state-space model of the PMDC motor can be expressed as:

$$\dot{\omega}_m = a \omega_m + b v_a - \frac{T_L}{J_m} \quad (7)$$

where

$$a = -\left(\frac{K_t K_b}{J_m R_a} + \frac{D_m}{J_m}\right) \quad (8)$$

$$b = \frac{K_t}{J_m R_a} \quad (9)$$

Let $a = a_0 + \Delta a$ and $b = b_0 + \Delta b$ where a_0, b_0 are the nominal values, and $\Delta a, \Delta b$ are the variations from parametric uncertainties. Consequently, Eq. (7) can be written as follows:

$$\dot{\omega}_m = a_0 \omega_m + b_0 v_a + d \quad (10)$$

where $d = \Delta a \omega_m + \Delta b v_a - T_L/J_m$ can be viewed as a disturbance from the parametric uncertainties and load torque.

Assumption 1: The disturbance $d(t)$ is bounded by a positive number Δ_{max} , i.e. $|d(t)| \leq \Delta_{max}$.

2.2. Finite-Time Control

Consider the nonlinear system given by:

$$\dot{s}(t) = f(t, s(t)), \quad s(0) = s_0 \quad (11)$$

where $s \in \mathbb{R}^n$ is the system state, and $f : \mathbb{R}^+ \times \mathcal{D} \rightarrow \mathbb{R}^n$ is locally Lipschitz continuous with $f(t, 0) = 0, t > 0$.

Lemma 1: [6] Assume that there exists a continuously differentiable, positive definite function $V(s)$ on an open neighborhood $\mathcal{U}_0$ of the origin such that:

$$\dot{V}(s) \leq -\gamma V^\alpha(s), \quad \forall s \in \mathcal{U}_0 \quad (12)$$

where $\gamma > 0$ and $0 < \alpha < 1$. Then, the origin is finite-time stable, and the reaching time is finite and satisfies the following inequality:

$$T_r(s_0) \leq \frac{1}{\gamma(1-\alpha)} V^{1-\alpha}(s_0) \quad \blacksquare \quad (13)$$

3. CONTROLLER DESIGN

3.1. Fractional Integral Terminal Sliding Mode Control

A FITSMC method is presented to achieve the speed tracking problem, i.e.,

$$\lim_{t \rightarrow \infty} \omega_m(t) = \omega_m^r(t), \quad (14)$$

where $\omega_m^r(t)$ is a reference trajectory. Therefore, let the tracking error be defined as:

$$e(t) = \omega_m^r(t) - \omega_m(t) \quad (15)$$

The FITSMC method is selected since the state-space model of the motor has the relative degree of one when the angular velocity is considered as the output.

The FITSMC is defined by the sliding variable as follows:

$$s(t) = e(t) + c \int_0^t e^{q/p}(\tau) d\tau, \quad c > 0 \quad (16)$$

where p and q are odd integers satisfying $p > q > 0$.

Remark 1: If the system is on the sliding surface: $s(t) = 0, \forall t \geq t_0$ with initial tracking error $e(t_0) = e_0$, the tracking error $e(t)$ converge to zero in finite settling time T_s given by [6]:

$$T_s(e_0) = \frac{|e_0|^{1-q/p}}{c(1-q/p)} \quad \blacksquare \quad (17)$$

To find the control v_a , the reaching law is selected as:

$$\dot{s} = -\varepsilon \operatorname{sgn}(s), \quad \varepsilon > 0 \quad (18)$$

This law is known as the constant-rate reaching law [7]. The parameter ε determines how fast it reaches the sliding surface.

By differentiating Eq. (16), we obtain:

$$\begin{aligned} \dot{s} &= \dot{e} + c e^{q/p} \\ &= [\dot{\omega}_m^r - \dot{\omega}_m] + c e^{q/p} \end{aligned} \quad (19)$$

By substituting $\dot{\omega}_m$ from Eq. (10) and $\dot{s}$ from Eq. (18) into Eq. (19), we have:

$$-\varepsilon \operatorname{sgn}(s) = [\dot{\omega}_m^r - (a_0\omega_m + b_0v_a + d)] + ce^{q/p} \quad (20)$$

Solving for v_a , we obtain:

$$v_a = \frac{1}{b_0} [\dot{\omega}_m^r - a_0\omega_m + ce^{q/p} + \varepsilon \operatorname{sgn}(s) - d] \quad (21)$$

Since all quantities on the right-hand side of Eq. (21) are known except the disturbance d , the control law in Eq. (21) cannot be implemented.

Consequently, by using the bound of d , the FITSMC is designed as follows:

$$\begin{aligned} v_a &= \frac{1}{b_0} [\dot{\omega}_m^r - a_0\omega_m + ce^{q/p} + \varepsilon \operatorname{sgn}(s) \\ &\quad + \Delta_{max} \operatorname{sgn}(s)] \\ &= \frac{1}{b_0} [\dot{\omega}_m^r - a_0\omega_m + ce^{q/p} + k \operatorname{sgn}(s)] \end{aligned} \quad (22)$$

where $k = \varepsilon + \Delta_{max}$ is the FITSMC gain.

3.2. Stability Analysis

Theorem 1: Under the assumptions of bounded disturbances and model uncertainties, the FITSMC given in Eq. (22) ensures that the tracking error converges to zero in a finite time. The closed-loop system is stable in the sense of Lyapunov, and the reaching time is finite, provided that the controller gain satisfies $k > \Delta_{max}$. ■

To prove Theorem 1, consider a candidate Lyapunov function:

$$V(s) = \frac{1}{2}s^2 \quad (23)$$

The derivative of $V(s)$ is given by

$$\dot{V} = s\dot{s} = s([\dot{\omega}_m^r - (a_0\omega_m + b_0v_a + d)] + ce^{q/p}) \quad (24)$$

By substituting v_a from Eq. (22) into Eq. (24), we have:

$$\dot{V} = s(-k \operatorname{sgn}(s) - d) = -k|s| - ds \quad (25)$$

From Assumption 1, $-ds \leq |ds| = |d||s| \leq \Delta_{max}|s|$. Consequently,

$$\dot{V} \leq -(k - \Delta_{max})|s| = -\gamma V^{1/2} \quad (26)$$

where $\gamma = \sqrt{2}(k - \Delta_{max})$. According to Lemma 1, $\gamma > 0$, or equivalently, the controller gain k must satisfy $k > \Delta_{max}$ to guarantee finite reaching time T_r , which is given by Eq. (13).

Consequently, since the settling time T_s , given by Eq. (17), is finite, the convergence time—equals the sum of T_r and T_s —is also finite. ■

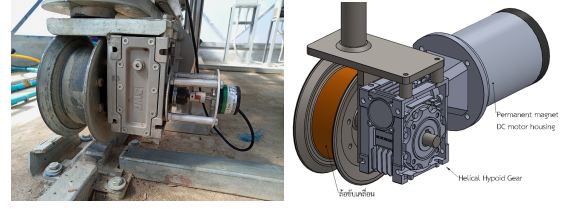


Fig. 2 DC motor mounted on the robot chassis along with its corresponding drive wheel.

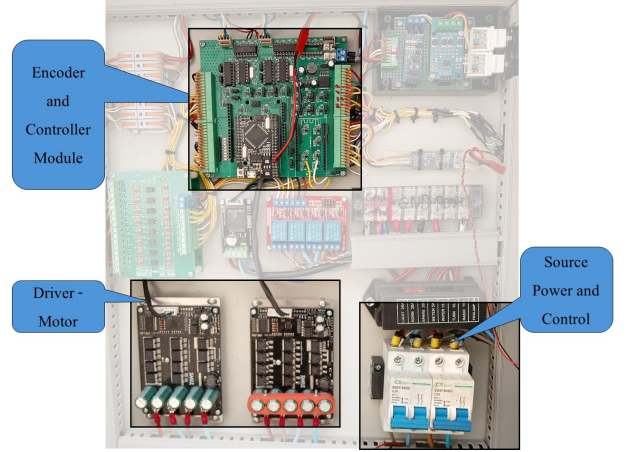


Fig. 3 Hardware of motor controller developed by our research group

4. SYSTEM IDENTIFICATION OF DC MOTOR

4.1. Hardware Design

Fig. 1 shows our phenotyping robot, which is designed to support high-precision motion control and modular sensor integration. The robot is designed to operate autonomously along predefined paths between crop rows, capturing plant phenotypic traits, such as height, leaf area, and canopy density, using onboard sensors.

Two DC motors, equipped with optical encoders, are used for locomotion, providing the position and velocity feedback necessary for closed-loop control. The frame is designed to minimize weight while maintaining sufficient structural rigidity for field conditions. Fig. 2 shows one of the DC motors mounted on the robot chassis along with its corresponding drive wheel.

The motor control system incorporates a feedback mechanism using encoders with differential signaling, ensuring robustness against signal interference. A custom-designed PCB facilitates this application, featuring electrical isolation between control and power drive circuits to minimize noise. The control board generates Pulse Width Modulation (PWM) signals, which are transmitted to the motor driver board for precise motor operation. The hardware layout is shown in Figure 3.

4.2. System Identification

4.2.1. Data collection

Since the wheel and motor do not have a predefined transfer function, it is necessary to determine the transfer function by measuring the motor speed in response to a

step-input voltage. The wheel is primarily driven by the motor, making the motor's transfer function a significant element. Initially, a first-order motor transfer function will be used to simplify the analysis, given that the input frequency is relatively low (< 1 kHz). Other dynamics will be treated as uncertainties, which will be addressed later to develop a robust control strategy.

Data collection for deriving the transfer function involves recording the motor's rotational speed as a function of voltage at 20 levels, sampled every 2 milliseconds over 60 seconds. This duration ensures the motor reaches steady-state conditions under no-load conditions for safety. The recorded data is then processed in MATLAB to observe the response. Signal processing techniques, such as Euler Backward differentiation and outlier detection and removal, are employed to filter out noise and remove anomalous data points. Finally, the cleaned data is used in the system identification process to extract a representative model for further use.

4.2.2. Motor Parameters

The transfer function is derived using the System Identification Toolbox in MATLAB. From the transfer function and the specification of the DC motor, the motor parameters are as follows: $K_t = 0.0712$; $K_b = 0.0712$; $R_a = 0.2022$; $J_m = 0.082$; $D_m = 0.569$.

5. RESULTS AND DISCUSSION

Three controllers are used to compare the effectiveness of controllers. The controllers are given by:

1. PID Controller:

$$v_a = K_p e + K_i \int_0^t e(\tau) d\tau + K_d \frac{de}{dt} \quad (27)$$

where $K_p = 32.79$, $K_i = 164.72$, and $K_d = -0.01$.

2. ISMC:

$$v_a = \frac{1}{b_0} [\dot{\omega}_m^r - a_0 \omega_m + ce + k \operatorname{sgn}(s)] \quad (28)$$

where $c = 400$, and $k = 20.5$.

3. FITSMC:

$$v_a = \frac{1}{b_0} [\dot{\omega}_m^r - a_0 \omega_m + ce^{q/p} + k \operatorname{sgn}(s)] \quad (29)$$

where $c = 400$, $k = 20.5$, $p = 3$, and $q = 1$.

All controller parameters are found by heuristic tuning.

The reference angular velocity is trapezoidal with a smoothing edge. Fig. 4 compares the angular velocity responses of the PID, ISMC, and FITSMC controllers with the desired reference trajectory. Tracking errors and absolute values of tracking errors are shown in Figs. (5) and (6).

All three controllers demonstrate effective tracking of the reference signal throughout the test period. As shown in Table 1, FITSMC and ISMC exhibit nearly identical and closely aligned responses with the reference signal, indicating high accuracy and strong control performance.

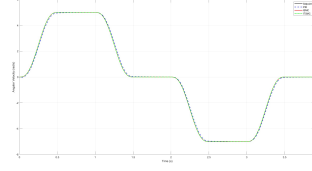


Fig. 4 Comparison of angular velocity tracking performance for PID, ISMC, and FITSMC controllers.

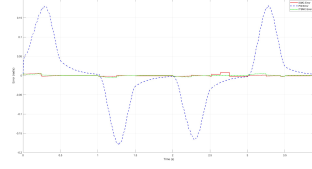


Fig. 5 Tracking errors for PID, ISMC, and FITSMC

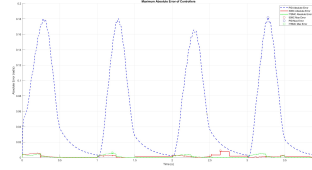


Fig. 6 Absolute values of tracking errors for PID, ISMC, and FITSMC.

However, the MAE of FITSMC is lower than that of ISMC, as shown in Table 2. Moreover, the FITSMC response is notably smoother. This makes FITSMC suitable for applications requiring precision and actuator protection, such as in agricultural phenotyping robots.

Table 1 Performance Metrics Comparison

Controller	ISE	IAE	ITAE
PID	0.041560	0.349776	0.420708
ISMC	0.000031	0.008725	0.005211
FITSMC	0.000032	0.009048	0.007985

Table 2 Maximum Absolute Error (MAE) Comparison

Controller	MAE (rad/s)
PID	0.181848
ISMC	0.008637
FITSMC	0.006115

6. CONCLUSION

This paper presents a FITSMC for precise motion control of a DC motor used in a phenotyping robot platform developed for green and sustainable agricultural applications. The proposed controller ensures finite-time convergence of the tracking error, offering enhanced robustness and fast response. To support controller design, a system identification process was conducted on a physical DC motor, and the resulting model was used to validate the control performance in a simulation environment. The simulation results confirm that the FITSMC approach achieves accurate angular velocity tracking under model uncertainties and external disturbances.

Future work will focus on real-time implementation of the proposed controller on the physical robot platform, along with field testing under realistic agricultural conditions and dynamic environments.

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